

Plot analysis (GPT-4o)

The plot shows the yield stress versus temperature for various Nb-Mo alloys, comparing the Maresca-Curtin model predictions with experimental data. Here's an analysis of the results:

Trend Identification

- **General Trend**: For all alloys, the yield stress decreases with increasing temperature, which is typical for metallic materials.
- **Model Predictions**: The Maresca-Curtin model predicts a smooth, nonlinear decrease in yield stress with temperature for each alloy.

Comparative Analysis

- **Nb95Mo5**: The model underestimates the yield stress compared to experimental data across the temperature range. The discrepancy increases at lower temperatures.
- **Nb91.5Mo8.5**: The model predictions are closer to experimental data, but still slightly underestimate yield stress, especially at lower temperatures.
- **Nb81Mo19**: The model aligns well with experimental data, showing only minor discrepancies.
- **Nb75Mo25**: The model overestimates the yield stress compared to experimental data, particularly at lower temperatures.

Accuracy of the Maresca-Curtin Model - The model generally captures the trend of decreasing yield stress with increasing temperature but shows varying accuracy across different alloy compositions.

- It performs best for Nb81Mo19, where the predictions closely match experimental data.
- The model is less accurate for Nb95Mo5 and Nb75Mo25, where it underestimates and overestimates the yield stress, respectively.

Discrepancies and Possible Reasons

- **Alloy Composition**: Variations in alloy composition can affect mechanical properties, and the model may not fully account for these effects.
- **Microstructural Factors**: Differences in grain size, phase distribution, or other microstructural features not captured by the model could lead to discrepancies.
- **Temperature Effects**: The model may not fully incorporate all temperature-dependent mechanisms affecting yield stress, such as dislocation movement or phase transformations.
- **Experimental Variability**: Experimental data can have inherent variability due to measurement errors or sample differences.

Suggestions for Improvement

- **Model Refinement**: Incorporating more detailed microstructural and compositional factors could improve model accuracy.
- **Parameter Calibration**: Adjusting model parameters based on a broader range of experimental data might enhance predictive capability.
- **Further Experiments**: Conducting additional experiments to explore the effects of microstructure and composition could provide insights for model adjustments.

Figure S6: Results of the analysis conducted by the multi-modal agent powered by GPT-4o for the plot shown in Figure 7(c).